RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

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Mysuru Road Bengaluru - 560 059

III Semester B.E. Regular / Supplementary Examination February/March 2026

ELECTRONICS AND COMMUNICATION ENGINEERING

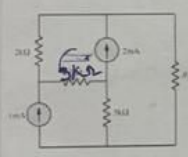
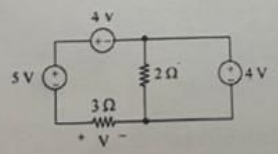
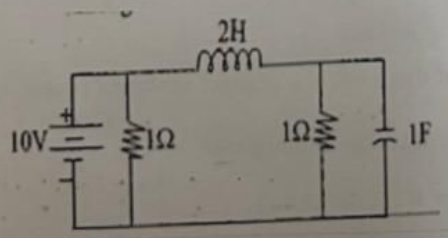
Network Analysis & Control Engineering

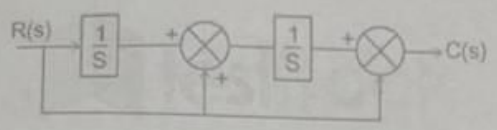
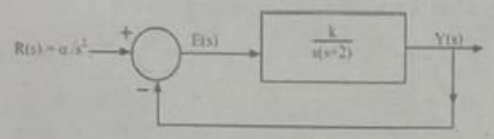
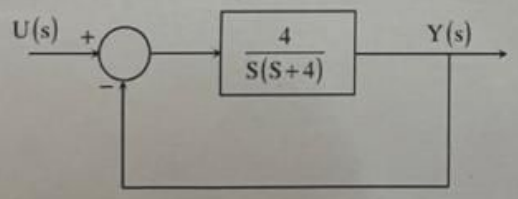
Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

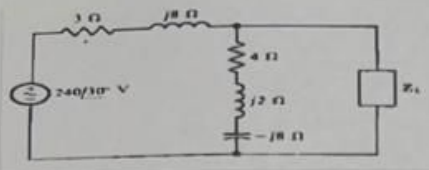
Part A

Question No	Question	M	CO	BT
1.1	Find R_L for maximum power transfer in the network shown in Figure  3K	02	1	4
1.2	The voltage V in Fig. is equal to _____ V. 	02	1	3
1.3	In Figure, at steady state, Find the current through the inductor. 	02	3	3
1.4	If the Laplace transform of the voltage across a capacitor of value 1/2 F is $Vc(S) = \frac{(s+1)}{(s^3+s^2+s+1)}$ Find the value of the current through the capacitor at $t = 0^+$.	02	1	3

1.5	For the block diagram shown in figure, the transfer function $\frac{C(s)}{R(s)}$ is equal to – 	02	2	3
1.6	Control Systems can be classified as open loop control systems and closed loop control systems based on the----------.	01	2	2
1.7	A closed loop system is shown in the figure where k>0 and α >0 . The steady state error due to a ramp input is given by 	02	4	4
1.8	The open-loop transfer function of a unity-feedback control system is $$G(S) = \frac{K}{s(s+2)}$$ For the peak overshoot of the closed-loop system to a unit step input to be 10%, the value of K is _____.	02	3	3
1.9	For the second order closed-loop system shown in the figure, the natural frequency (in rad/s) is _____. 	01	1	1
1.10	The open-loop transfer function of a unity-feedback control system is $$G(s) = \frac{K}{s^2+5s+5}$$ The value of K at the breakaway point of the feedback control system's root-locus plot is _____.	02	4	4
•1.11	How is the gain of the system made half of the gain margin?	02	4	5

Part B

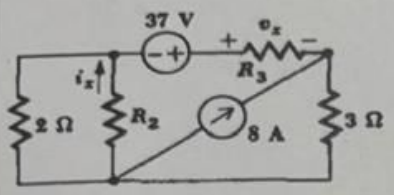
Question No	Question	M	CO	BT
•2a	In the circuit of Fig , Design load impedance Z_L that absorbs maximum average power and determine the maximum Power.	10	1	5



Using mesh analysis, find i_x and V_x for the network shown in Fig with

$R_2 = 4\Omega$ and $R_3 = 5\Omega$.

2b

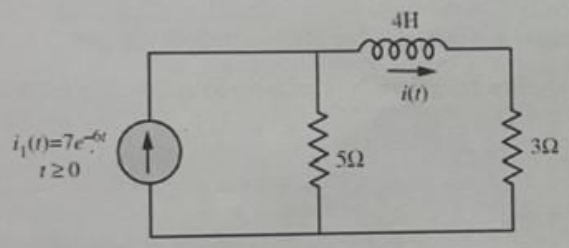


06 1 3

Find i(t), for the circuit of Figure, when, $i_1(t) = 7e^{-6t}\ A\ for\ t \geq 0\ and\ i(0) = 0.$

Also find $i(\infty)$.

3a



08 2 3

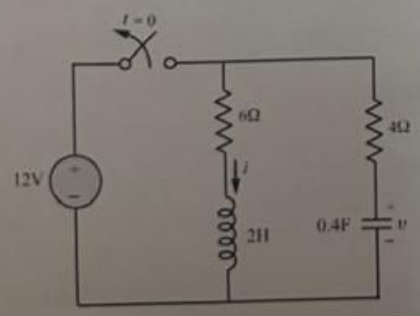
For the circuit shown in Figure, find:

a) $i(0+)$ and $v(0+)$

b) $\frac{di(0+)}{dt}$ *and* $\frac{dv(0+)}{dt}$

c) $i(\infty)$ and $v(\infty)$.

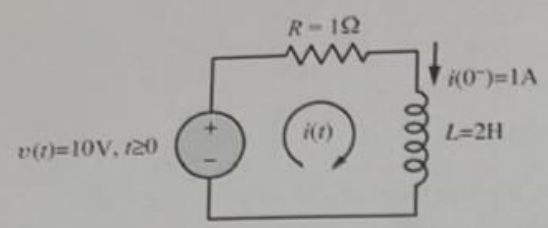
3b



08 4 4

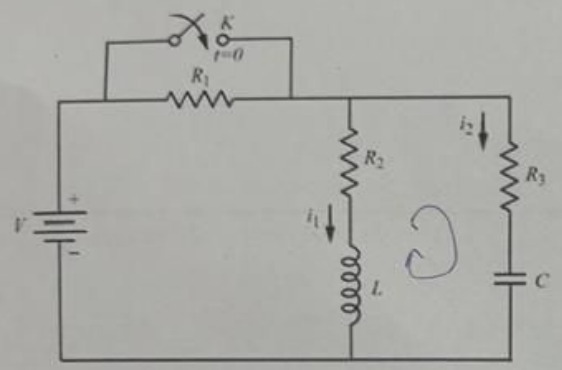
OR

•4a Refer the circuit shown in Fig. Find $i(0)\ and\ i(\infty)$ using initial and final value theorems. also find current i(t) for t>0. 08 2 4



For the network shown in Figure below with switch K open, a steady-state is reached. The circuit parameters are
$R_1 = 10\Omega$, $R_2 = 20\Omega$, $R_3 = 20\Omega$, L = 1 H and C = 1F. Take V = 100 volts. The switch is closed at t = 0.

4b



08 3 4

(i) Write the integro-differential equation after the switch is closed.

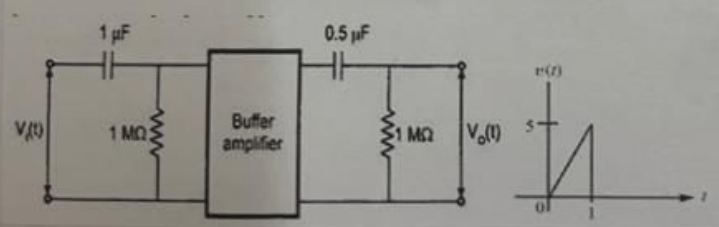
(ii) Find the voltage across before the switch is closed and give its polarity.

(iii) Find i_1 and i_2 at t = 0+.

(iv) Find $\frac{di_1}{dt}$ and $\frac{di_2}{dt}$ at t = 0+.

5a

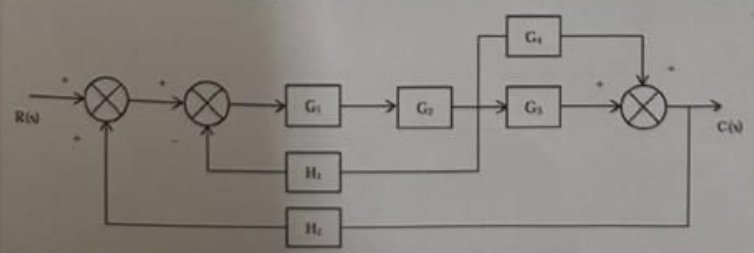
Find $Vo(S)$ $Vi(S)$ for the electrical circuit shown in Figure . Also, find the output of the system if unit step input is applied



08 2 5

5b

Find the transfer function $\frac{C(s)}{R(s)}$ for the block diagram shown in below figure.



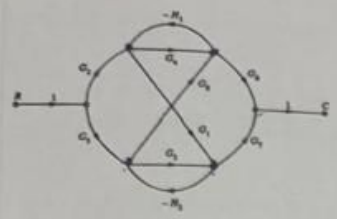
08 2 3

OR

6a

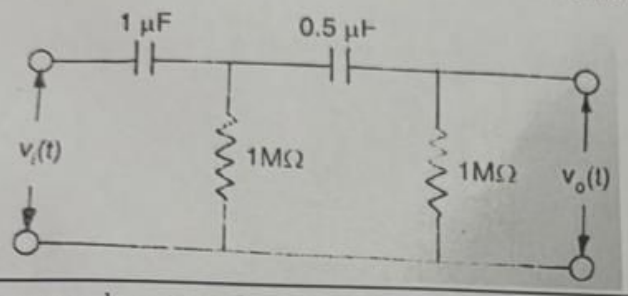
Consider the control system whose signal flow graph is shown below. Determine the system transfer function using Mason's formula.

08 3 3



Derive the transfer function for the electric network shown.

6b

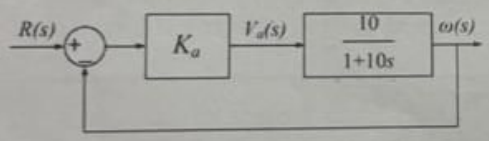


08 2 3

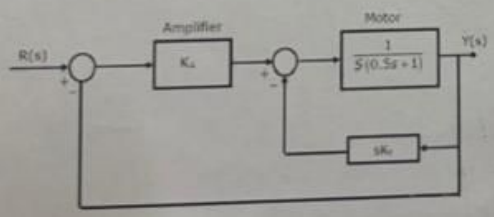
The open-loop transfer function of a motor is given as

$$\frac{\omega(s)}{V_a(s)} = \frac{10}{1+10s}$$

7a When connected in feedback as shown below, find the approximate value of K_a that will reduce the time constant of the closed loop system by one hundred times as compared to that of the open-loop system. 06 2 2



Block diagram model of a position control system is shown in figure.



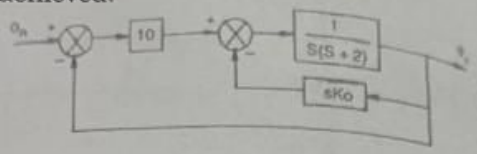
7b 10 3 4

(i) In absence of derivative feedback (K_t=0), determine damping ratio of the system for amplifier gain K_A=5. Also find the steady state error to unit ramp input.

(ii) Find suitable values of the parameters K_A and K_t so that damping ratio of the system is increased to 0.7 without affecting the steady state error as obtained in part (i).

OR

8a A servo system for the positional control of a rotatable mass is stabilised by viscous friction damping which is three quarters of that is needed foe critical damping. The undamped natural frequency of the system is 12 Hz. Derive an expression for the output of the system, if the input control is suddenly moved to a new position, being initially at rest. Hence find the maximum overshoot. 08 4 4

8b	The system shown in Fig employs derivative of feedback in addition to unity feedback. (i) Find the value of constant K_0 so that the damping ratio of the system is 0.7. With this calculated value of K_0 and unit ramp input, find the value of steady state error. (ii) If $K_0 = 0$, calculate the value of damping ratio and undamped frequency of oscillations. What will be the steady state error resulting from unit ramp input. (iii) The value of steady state error is required to be 0.2 radians with derivative feedback and the damping ratio maintained at 0.7. How can this be achieved.  	08	3	3
9a	Plot the Bode magnitude and phase diadrams for the open loop transfer function $G(S)H(S) = \frac{100(S+2)}{S(S+4)(S+5)}$ Discuss the stability of the closed loop system.	10	2	5
9b	With the help of Routh–Hurwitz criterion and example, explain how the stability of a control system can be determined without finding the roots of the characteristic equation.	06	4	1
	OR			
10a	Draw the Root Locus and determine the stability of the system with open loop transfer function is $G(S)H(S) = \frac{k}{s(s+2)(s+8)}$	08	3	4
10b	**The characteristic polynomial of a system is** $s^7 + 5s^6 + 9s^5 + 9s^4 + 4s^3 + 20s^2 + 36s + 36 = 0$. Determine the location of roots on the s-plane and hence the stability of the system.	08	4	4